

SOC Internship – Day 23 Notes

Active Directory Security

Objective

Understand how SOC analysts monitor and secure Active Directory environments against cyber attacks.

1. What is Active Directory?

Active Directory (AD) is Microsoft's centralized identity and access management system used in enterprise environments.

Main Functions

- User authentication
- Authorization
- Group management
- Access control
- Security policy management

Simple Explanation

Active Directory helps organizations manage users, computers, passwords, and permissions from one central system.

2. Why Active Directory is Important

Active Directory is important because it provides:

- Centralized authentication
- User and group management
- Access control
- Enterprise security management

Benefits

- Easier administration
- Improved security
- Centralized monitoring
- Better user management
- Secure access to company resources

3. Common Active Directory Attacks

a) Password Spraying

Attackers try common passwords against many accounts.

Example

Trying:

- Welcome123
- Password@123

Against multiple users.

Goal

Gain access without triggering account lockouts.

b) Kerberoasting

Attackers request Kerberos service tickets and crack them offline to recover passwords.

Target

Service accounts with weak passwords.

Risk

Privilege escalation and lateral movement.

c) Pass-the-Hash (PtH)

Attackers use stolen password hashes instead of actual passwords to authenticate.

Risk

Unauthorized access to systems.

d) Privilege Escalation

Attackers gain higher privileges such as Domain Admin access.

Risk

Complete control over the network.

e) Account Compromise

User accounts are stolen using phishing, malware, or credential theft.

Risk

Data theft and unauthorized access.

4. Important AD Indicators

SOC analysts monitor several indicators to detect attacks:

- Multiple failed logins
- Privileged group changes
- Suspicious account creation
- Unusual admin logins

Why These Indicators Matter

These activities may indicate:

- Brute force attacks
- Insider threats
- Privilege escalation

- Compromised accounts

5. Example Investigation

Alert

Multiple failed login attempts detected.

Findings

User added to the **Domain Admins** group.

Analysis

Potential privilege escalation attack.

Investigation Steps

1. Check login source IP
2. Verify account owner
3. Review recent admin actions
4. Analyze authentication logs
5. Confirm if privilege change was authorized

6. SOC Monitoring Areas

SOC teams continuously monitor:

Domain Controller Logs

Track authentication and system events.

User Authentication Events

Monitor login successes and failures.

Group Membership Changes

Detect unauthorized admin privilege changes.

Privileged Account Activity

Monitor admin account behavior.

7. SOC Team Responsibilities

L1 Analyst

- Monitor authentication alerts
- Escalate suspicious events

L2 Analyst

- Investigate suspicious activity
- Analyze attack indicators

L3 Analyst

- Improve AD detection rules
- Enhance incident response processes

8. Real SOC Workflow

Domain Controller Logs

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SIEM Correlation

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Privilege Escalation Alert

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SOC Investigation

Explanation

- Logs are collected from domain controllers
- SIEM correlates suspicious activities
- Alert is generated
- SOC team investigates the incident

Class Activity Answer

Scenario

- User added to Domain Admins
- Multiple failed logins observed

Why is this suspicious?

Answer

This is suspicious because:

- Multiple failed logins may indicate brute force attempts
- Adding a user to Domain Admins increases privileges
- Attackers often escalate privileges after compromising an account

Possible Threat

Privilege Escalation Attack

Assignment Answers

1. Explain Active Directory

Active Directory is Microsoft's centralized system used to manage users, authentication, permissions, and access control within enterprise networks.

2. Research Common AD Attacks

Common AD Attacks

- Password Spraying
- Kerberoasting
- Pass-the-Hash
- Privilege Escalation
- Account Compromise

Impact

These attacks can lead to unauthorized access, data theft, and complete domain compromise.

3. Analyze Privilege Escalation Scenario

Scenario Analysis

An attacker first attempts multiple logins. After gaining access, the attacker adds a user account to the Domain Admins group to gain elevated privileges.

Indicators

- Failed login attempts
- Admin group membership change
- Suspicious authentication activity

SOC Response

- Investigate logs
- Disable compromised account
- Remove unauthorized privileges
- Reset passwords
- Monitor for further malicious activity